INCH-POUND
A-A-50532A(MC)
October 8, 2021
SUPERSEDING
A-A-50532
September 19, 1990

COMMERCIAL ITEM DESCRIPTION

CLOTH, SERGE: POLYESTER/WOOL

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers the requirements for one type of polyester/wool serge cloth.

2. SALIENT CHARACTERISTICS.

2.1 Material description. The cloth shall be a 2/2 right hand twill weave using a spun polyester and wool fiber blended into a single yarn and woven with a continuous filament polyester yarn into an end-and-end and pick-and-pick arrangement in both warp and filling of the body of fabric.

2.2 Color. The color of the finished cloth shall be as specified in the contract or purchase order (see 6.4). The finished cloth shall present a solid color with no more heatheriness than the standard sample.

2.2.1 Visual shade matching. The color and appearance of the material shall match the standard sample when viewed using AATCC EP9, Visual Assessment of Color Difference of Textiles, option C (see 6.5), with sources simulating artificial daylight D65 illuminant (see 6.5.1) with a color temperature of 6500±200 Kelvin (K) illumination of 100±20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856±200K.

2.3 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested in accordance with the test methods cited in table I.

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FSC 8305

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TABLE I. Physical requirements.

| Characteristic | Requirements | Test Method |
|---|---|---|
| Weave | 2 up/2 down right-hand twill | Visual [1/] |
| Fiber content (percent) ±3 percent | | |
| Polyester | 80 | AATCC TM20A (see 6.3) |
| Wool | 20 | |
| Wool grade (min.) | 62's or finer | U.S. Standard for Grades of Wool |
| Weight, oz/yd² | 6.3 to 6.6 | ASTM D3776/D3776M, option C |
| Yarns per inch (min.) | | |
| Warp | 65 | ASTM D3775 |
| Filling | 55 | |
| Breaking strength, pounds (min.) | | |
| Warp | 160 | ASTM D5034 |
| Filling | 140 | |
| Colorfastness (min.) | | |
| Crocking (wet and dry) | 3.5 | AATCC TM8 [2/] |
| Perspiration (acid and alkaline) | | |
| Color change and staining | 4 | AATCC TM15 [3/] |
| Lightfastness (after 40 AFU or 170 kJ/[$m^2$nm] at 420 nm) | 4 | AATCC TM16.3 [4/] |
| Wet dry cleaning | | |
| Color change and staining | 4 | AATCC TM132 [3/] |
| Dimensional change, (after 3 cycles) percent (max.) | | |
| Warp | ±2.5 | AATCC TM135 (1)(V)(Aiii) [5/] |
| Filling | ±2.5 | |
| pH | 5.5 to 8.5 | AATCC TM81 |
| Pilling | 3 | ASTM D3511/D3511M |
| Toxicity | 2.4 | 2.4.1 |

FOOTNOTES:

1/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

2/ Rated using AATCC EP8, 9-Step Chromatic Transference Scale.

3/ Rated using AATCC EP 1, Gray Scale for Color Change, and AATCC EP2, Gray Scale for Staining.

4/ Rated using AATCC EP 1.

5/ Specimens shall be subjected to three complete cycles (wash and dry) prior to determination of dimensional change.

2.4 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 2.4.1. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used. When specified (see 6.4), the contractor shall furnish information that certifies that the finished product is composed of materials that have been safely used commercially or provide sufficient toxicity data to show compatibility with prolonged, direct skin contact.



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2.4.1 Toxicity assessment. When specified (see 6.4), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of the studies indicate the cloth is not a sensitizer or irritant, a human Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure in "Hayes' Principles and Methods of Toxicology." If the toxicity requirement can be demonstrated with historical data on the finished treatments used, then toxicity testing may not be required.

2.5 Labels. Each roll of finished cloth shall be labeled or ticked for fiber content in accordance with 16 CFR § 303, Rules and Regulations Under the Textile Fiber Products Identification Act.

2.6 Workmanship. The finished cloth shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the contractor's own quality assurance standards and the quality assurance standards defined by the technical data in the bid package.

3. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The finished cloth provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. This finished cloth must have been sold to the commercial market or the government.

4.3 Visual examination. The finished cloth shall be examined for the following defects as defined in ASTM D3990, Standard Terminology Relating to Fabric Defects:

a. Cloth – any spot or stain more than 1 inch in its longest dimension; any hole, cut, or tear; a missing end or pick; thick or thin areas; overall uncleanliness; a crease or hard crease; slubs, knots, slugs, or gout; and shade bar, streaks, or uneven dying. Water repellent treatment not uniform.

b. Label – label missing, incorrect, or illegible. Required information missing from the label.

c. Packaging – not packaged in accordance with the contract or purchase order (see 6.4).

4.4 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 6.4).

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Intended use. This cloth is intended to be used for military uniform components.

6.2 Source of documents.

6.2.1 American Association of Textile Chemists and Colorists (AATCC). AATCC test methods are available from www.aatcc.org or P.O. Box 12215, Research Triangle Park, NC 27709.

6.2.2 ASTM International. ASTM standards are available from www.astm.org or ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959.

6.2.3 Code of Federal Regulations (CFR). The CFR is available from www.ecfr.gov or the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20401.

6.2.4 Hayes' Principles and Methods of Toxicology. Hayes' Principles and Methods of Toxicology is available from www.crcpress.com or may be available from other retailers.

6.2.5 United States Standards for Grades of Wool. United States Standards for Grades of Wool are available from www.ams.usda.gov.



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6.3 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirements.

6.4 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Color required (see 2.2).
c. When toxicity testing is required (see 2.4 and 2.4.1).
d. Product conformance provisions (see 4.1).
e. Packaging requirements (see 4.3.c and 5).
f. Acceptance criteria provisions (see 4.4).

6.5 Visual shade matching. In 2019, option A of AATCC EP9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35±5-degree angle relative to the horizontal. The observer will view the specimens at a 90-degree angle, relative to the plane of the specimens.

6.5.1 Use of D75 illuminant. The use of D75 illuminant with a color temperature of 7500±200 Kelvin (K) illumination of 100±20 foot candles in lieu of the specified D65 illuminant is permitted.

6.6 Key words.

Uniform

CONCLUDING MATERIAL

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions to this Commercial Item Description until further notice.

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – MC
(Project 8305-2021-024)



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